



5-HT_{1A} receptor blockade potentiates the subjective effects of DMT

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Abstract

Introduction: Serotonergic psychedelics are being investigated in the treatment of various disorders and in the improvement of well-being. Evidence suggests that their subjective effects may play a role in long-term behavioral outcomes. The subjective effects are mediated by 5-HT_{2A} receptor agonism, but the 5-HT_{1A} receptor may also play a role in the subjective effects. This study elucidates the effects of 5-HT_{1A} receptor blockade using pindolol pre-treatment on the subjective effects induced by *N,N*-dimethyltryptamine (DMT).

Methods: In a double-blind, randomized, placebo-controlled within-subjects design, 12 (10 males, 2 females) experienced hallucinogen-using participants received a sub-hallucinogenic dose of intravenous DMT fumarate, 0.1 mg/kg, after pre-treatment with 30 mg oral racemic pindolol. Subjective effects were measured using the Hallucinogen Rating Scale.

Results: Pindolol pre-treatment increased DMT-induced subjective effects with a moderate effect size ($M=0.514$). Blood pressure and mean arterial pressure also increased with pindolol pre-treatment at 2 minutes following DMT administration, but heart rate was not affected.

Conclusions: 5-HT_{1A} receptor blockade results in a global intensification of DMT-induced subjective effects, suggesting a functional role of 5-HT_{1A} receptor action in the mechanism of psychedelic-induced subjective effects.

Keywords

psychedelics, subjective effects, DMT, serotonin receptor, 5-HT_{1A}, 5-HT_{2A}

Introduction

Serotonergic psychedelic compounds are increasingly being studied as therapeutics in various clinical populations and to enhance well-being in healthy participants (Goodwin et al., 2023; Griffiths et al., 2016; Raison et al., 2022). Evidence suggest that the psychological changes elicited by the acute subjective experience may play an integral role in the long-term therapeutic and well-being outcomes (Yaden and Griffiths, 2021). For example, in a trial investigating the efficacy of the serotonergic psychedelic psilocybin in the treatment of treatment-resistant depression, researchers found that the quality of the experience significantly predicted therapeutic outcomes. Specifically, increased oceanic boundlessness (associated with dissolution of the self and indicative of more “mystical” type and insightful experiences), and less dread of ego dissolution (associated with more anxiety-like experiences) were associated with long-term benefit in participants (Roseman et al., 2017; Yaden et al., 2024).

The acute psychedelic experience is characterized by perceptual disturbances, including visual and auditory hallucinations, intense affective responses, a dissolution of the self, and psychological insight (Metastasio et al., 2025). These effects are purported to occur primarily through agonism at the 5-HT_{2A} receptor (Madsen et al., 2019; López-Giménez and González-Maeso, 2018) as pretreatment with the 5-HT_{2A} receptor antagonist ketanserin or the dopamine/5-HT_{2A} antagonist risperidone can fully block the subjective effects of psilocybin (Carter et al., 2005; Komater et al., 2012; Vollenweider et al., 1998).

However, serotonergic psychedelics (tryptamine and ergoline, but generally not phenethylamine classes) also show

significant affinity for the 5-HT_{1A} receptor as well (Kelmendi et al., 2022). Despite this, the contributions of the 5-HT_{1A} receptor in the psychedelic experience have not been comprehensively characterized. Evidence suggests that the 5-HT_{1A} receptor, either directly or indirectly through functional interaction with 5-HT_{2A} receptors, may play a role in the acute subjective effects induced by psychedelics (Halberstadt et al., 2011; Krebs and Geyer, 1994; Krebs-Thomson and Geyer, 1998). Preclinical evidence is mixed, where some studies suggests an inhibitory effect of 5-HT_{1A} receptor agonism with psychedelic administration others implicate a broader modulatory role (Berendsen and Broekkamp, 1990; Canal and Morgan, 2012; Fox et al., 2010; Glatfelter et al., 2024; Krebs-Thomson and Geyer, 1998). In a study examining the effect of 5-HT_{1A} agonist and antagonist coadministration on the effect of LSD in rodent test of drug discrimination, authors found that 5-HT_{1A} agonism with buspirone increased LSD induced increases in drug discrimination, where administration

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of 5-HT_{1A} antagonist WAY-100635 significantly attenuated drug discrimination (Reissig et al., 2005). In another paper, administration of 5-HT_{1A} antagonist WAY-100635 did not result in attenuation of the drug discrimination response with psilocybin (Winter et al., 2007). Such findings make clear a modulatory role of 5-HT_{1A} receptor activation, though the nature of the modulation is difficult to interpret and makes translation to human models difficult.

Such inconsistent 5-HT_{1A} receptor effects on behavior prompt more robust investigation, a point that is underscored by the fact that evidence of these effects in humans remains limited. One of two studies analyzing 5-HT_{1A} receptor agonism on psychedelic effects in humans administered the 5-HT_{1A} receptor agonist buspirone (20 mg), 60 minutes before the administration of psilocybin, and found that 5-HT_{1A} receptor agonism significantly reduced the intensity of various aspects of the subjective experience, particularly visual effects and oceanic boundlessness (Pokorny et al., 2016).

In an earlier report, Strassman (Strassman, 1996) conducted a clinical trial assessing the effects of 5-HT_{1A} receptor blockade with 90 minute pre-treatment of pindolol (30 mg), on the subjective effects of intravenous (IV) DMT fumarate (0.1 mg/kg). Though this trial was the first examination of the effects of the 5-HT_{1A} receptor on psychedelic-induced subjective effects, data were not quantitatively reported. Anecdotally, Strassman (1996) reported: "volunteers found that pindolol pre-treatment enhanced DMT effects by two to three times." Here, we quantitatively analyze the results from that study to elucidate the effects of 5-HT_{1A} receptor blockade on the subjective experience of DMT.

Methods

Participants

Ten male and two female experienced hallucinogen users were medically and psychiatrically screened (Strassman et al., 1994; Strassman and Qualls, 1994) to exclude those with current medical or psychiatric disorders, or a history of psychosis. All participants were psychedelic experienced, and the average age ($\pm$ SD) was 44.8 $\pm$ 1.6 years (range 39–59 years). Following rigorous medical and psychiatric screening described in detail in the original text (Strassman, 1996), participants consented to inclusion in the study. Participants were dosed from September of 1993 to February of 1994, and treatments were separated by at least 1 week for males and by at least 1 month for females. Females were studied during the early follicular phase of their menstrual cycle to control for the neuroendocrine milieu.

Study design

The data used herein were from a randomized, double-blind, within-subjects placebo-controlled trial of four treatments: (1) DMT (0.1 mg/kg IV) with placebo pre-treatment at 90 minutes, (2) DMT (0.1 mg/kg) with pindolol (30 mg) pre-treatment at 90 minutes, (3) Pindolol (30 mg) with placebo pre-treatment at 90 minutes and, (4) Placebo with placebo pre-treatment at 90 minutes. In all studies took place in the morning.

This study was part of a series of trials conducted previously by Strassman (1996) that took place in the inpatient unit of the University of New Mexico Hospital General Clinical Research

Center. To determine the optimal dose of DMT to combine with 30 mg racemic oral pindolol, three subjects who had been involved in previous DMT experiments were studied in a non-blind manner. Having previously participated in other DMT studies, subjects were familiar with the effects of intravenous (IV) DMT doses ranging from 0.05 to 0.4 mg/kg.

These pilot subjects first received 0.05 mg/kg intravenous (IV) DMT, a "just noticeable" dose, with and without pindolol, and subjective and cardiovascular responses were compared. As 0.05 mg/kg DMT plus pindolol was well-tolerated, subjects returned for a similar experiment with 0.1 mg/kg DMT, which was also well-tolerated. However, when 0.2 mg/kg DMT was combined with pindolol, one of the two subjects in this arm of the pilot study briefly became delirious and hypertensive. This particular subject had previously tolerated 0.4 mg/kg DMT, the ceiling dose. Since it appeared that pindolol pre-treatment more than doubled subjective and blood pressure responses to DMT, 0.1 mg/kg DMT was chosen.

If a subject was new to the DMT studies, they first received non-blind "screening" doses of 0.05 and 0.4 mg/kg IV DMT on consecutive days in the inpatient unit of the General Clinical Research Center (GCRC) of the University of New Mexico Hospital. This was in order to acquaint themselves with barely noticeable and maximal subjective and cardiovascular effects of the drug.

The present analysis of subjective effects focused on the comparison between DMT with placebo pre-treatment versus DMT with pindolol pre-treatment to assess the effects of 5-HT_{1A} blockade on DMT-induced subjective effects. An analysis of the cardiovascular effects included all four groups.

Drugs

N,N-Dimethyltryptamine fumarate was synthesized by David E. Nichols, Department of Medicinal Chemistry, Purdue University, West Lafayette, IN. It was dissolved in sterile saline at a concentration of 20 mg/ml and administered (0.1 mg, IV). DMT was infused into a forearm vein via an indwelling catheter over 30 seconds and the line was flushed with sterile saline for an additional 15 seconds. Placebo DMT was USP sterile saline.

Racemic pindolol was donated by Sandoz Pharma, Ltd, East Hanover, NJ. Pindolol was pulverized and placed into gelatin capsules and administered in 30 mg PO. Placebo pindolol was USP lactose placed into identical capsules.

Measures

Subjective effects outcome measures. After the acute psychological effects of DMT had resolved (30–45 minutes post-injection), subjects were interviewed in an open-ended manner for qualitative assessment of psychological effects, which were quantified using the Hallucinogen Rating Scale (HRS). The HRS is a 125-item, self-administered instrument. Items are placed into a priori "clinical clusters" corresponding to six subscales. The Somaesthesia subscale illustrates effects on interoceptive and somatic cues, sample scale questions included "(Did you feel) a rush?" and "(Did you feel) an electric/tingling feeling". The Affect subscale quantifies mood and affect; sample scale questions included "(Did you feel) anxious?" and "(Did you feel) excited?".

The Perception subscale addressed visual, auditory, olfactory, tactile, and gustatory perception of the experience. Sample questions included “(Did you feel) flushed?” and “(Did you hear) a sound or sounds accompanying the experience?”. The Cognition subscale probed thought processes and content and included questions such as “(Please describe visions/hallucinations) sense of speed” and “(Did you have) new thoughts and insights?”. The volition subscale asked about the ability to willfully interact with one’s mental and physical self. Sample questions included “(Did you experience) a change in effort of breathing?” and “(Were you) able to let go?”. Finally, a global intensity measure included questions such as “(Specify the) amount of time between when the drug was administered and an effect” and “(Was there) a waxing and waning of the experience?” (Strassman et al. 1994).

Cardiovascular measures. Heart rate and blood pressure were measured at -30, -1, 2, 5, 10, 15, 30, and 60 minutes, using an automatic blood pressure cuff for a 60-minute period following DMT administration and reported as individual change from baseline. Baseline values for cardiovascular measures were obtained by averaging -30- and -1-minute values. The analysis of these measures was descriptive without inferential statistics.

Statistical analysis

We implemented a Bayesian hierarchical linear model using the *rethinking* package in R statistical software (R Core Team, 2025). This package is a wrapper for Stan using Hamiltonian Monte Carlo sampling for Bayesian models.

Data standardization. Prior to analysis, HRS subscale scores were z-score standardized using the transformation:

$$(x - \mu) / \sigma$$

where μ and σ represent the mean and empirical sample standard deviation, respectively, computed across both treatment groups within a subscale. Standardization serves multiple critical functions: (1) It places all subscales on a common metric enabling direct comparison of effect magnitudes across domains with different raw scores; (2) it stabilizes the numerical properties of the Hamiltonian Monte Carlo sampling algorithm by ensuring predictors and outcomes are on similar scales; (3) it yields regression coefficients expressed in standardized units (i.e., per-SD changes in the outcome); and (4) it simplifies prior specification. The original scale means and standard deviations were preserved as attributes to enable back-transformation of posterior estimates to raw HRS units for interpretability.

The data were structured in long format with six rows for each subscale per subject in either treatment condition (DMT with placebo pretreatment and DMT with pindolol pretreatment). The model estimated subscale-specific treatment effects while accounting for between-subject variability and subscale-specific baselines. The likelihood and linear model were specified as:

$$y_{ij} \sim \text{Normal}(\mu_{ij}, \sigma), \mu_{ij} = a + a_i + a_j + b_j \text{cond}_{ij},$$

where

- y_{ij} = HRS score for participant i on subscale j
- α = grand intercept
- $\alpha_i \sim \text{Normal}(0, \sigma_s)$ = participant – specific random intercept
- α_j = subscale – specific intercept
- β_j = subscale – specific treatment slope
- $\text{cond}_{ij} \in \{-0.5, +0.5\}$ codes DMT + placebo versus DMT + pindolol pretreatment

This parameterization produces one treatment effect per subscale while sharing information across subjects and allowing each subscale to have its own baseline level.

Prior specification. Weakly informative priors were selected to regularize estimates without strongly constraining inference. Priors were:

$$\begin{aligned} a & \sim \mathcal{N}(0, 1) \\ a_i & \sim \mathcal{N}(0, \hat{\sigma}_s) \\ a_j & \sim \mathcal{N}(0, 1) \\ \beta_j & \sim \mathcal{N}(0, 1) \\ \sigma & \sim \text{Exponential}(1) \\ \sigma_s & \sim \text{Exponential}(1) \end{aligned}$$

- **Grand intercept (α):** Normal(0,1) centers the overall mean at zero with plausible variation of ± 2 SD on the standardized scale.
- **Subscale baselines (α_j):** Normal(0,1) reflects that subscale means should vary modestly around the pooled mean after z-scoring.
- **Subscale treatment effects (β_j):** Normal(0,1) is a weakly informative prior expressing skepticism toward extremely large standardized effects ($|d| > \sim 2$) while remaining agnostic about direction.
- **Subject-level SD (σ_s) and residual SD (σ):** Exponential(1) provides mild regularization, shrinking away implausibly large variance estimates without affecting moderate-sized values.

The exponential priors for variance components prevent implausibly large variance estimates while remaining weakly informative, and the Normal(0,1) priors on effects are well-calibrated for standardized outcomes.

Posterior sampling. The model was fitted using Hamiltonian Monte Carlo via Stan, with four independent chains each run for 4000 iterations (including warmup). Convergence was assessed using standard diagnostics, including $\hat{R}$ statistics and effective sample size.

Posterior inference and interpretation. We obtained posterior samples for each β_j parameter, representing the subscale-specific treatment effects. These coefficients quantify the expected change in standardized HRS scores when moving from DMT-only to DMT with pindolol pre-treatment conditions. Because

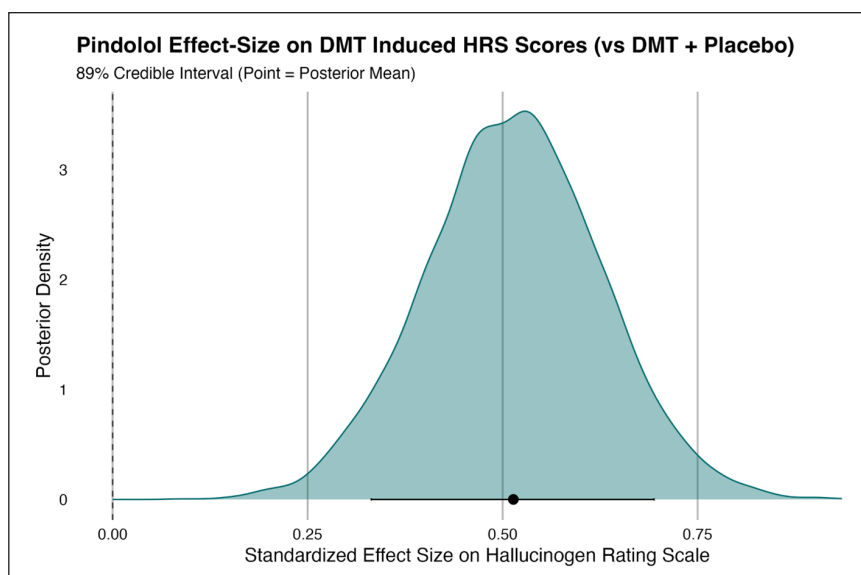


Figure 1. Mean standardized effect size of pindolol pre-treatment on DMT subjective effects.

Posterior distribution of mean standardized effect of 60-minute pindolol pre-treatment on DMT-induced subjective effects. Posterior mean is denoted with a point at $M=0.514$. DMT: *N,N*-dimethyltryptamine.

outcomes are z-scored, β_j values are directly interpretable as standardized effect sizes.

For each subscale-specific effect, we computed:

- **Posterior mean:** the expected effect size.
- **89% credible interval (CrI):** the range containing 89% of the posterior probability mass, representing plausible values given the data and model.
- **Probability of direction:** the proportion of posterior samples with $\beta_j > 0$, quantifying certainty about effect direction.

Posterior samples were *back-transformed to the original HRS scale* by multiplying standardized effects by the subscale-specific standard deviations preserved during preprocessing. This yields estimates in raw HRS score units, facilitating interpretation relative to the questionnaire's measurement properties.

A *global treatment effect* was computed by averaging across the six subscale-specific β_j values for each posterior sample, providing an overall summary of pindolol's impact on the DMT experience. Both standardized and raw-scale estimates with 89% CrIs were reported for this global effect.

Raw Cohen's *d*. Effect sizes were expressed as Cohen's *d*, computed as the difference between group means divided by the pooled standard deviation of the two groups.

$$d = (\bar{X}_1 - \bar{X}_2) / \sqrt{\left(s^2 \frac{1}{2} + s^2 \frac{2}{2}\right) / 2}$$

Software and reproducibility. All data pre-processing and analysis were conducted in R statistical software using custom-written analysis scripts (version 4.4.3; R Core Team, 2025). Visualization was conducted in both R and in Matlab software using

custom-written scripts (version R2024b; The MathWorks, Inc., 2024). All analysis and visualization code is accessible at the Open Source Development Network (Zahid, 2025).

Results

Pindolol effect on DMT-induced subjective experience

Pindolol pre-treatment increased DMT subjective effects as measured through the HRS. The posterior mean standardized effect size [89% CrI] was 0.514 [0.331, 0.694] (Figure 1) while the mean raw score difference was 0.297 [0.190, 0.403] (Supplemental Figure 1).

Individual subscale mean standardized effect sizes are as follows: (1) Intensity: mean=0.654, 89% CrI [0.219, 1.082], with a posterior probability positive of $P(>0)=0.99$; (2) Somaesthesia: mean=0.523, 89% CrI [0.092, 0.956], with a posterior probability positive of $P(>0)=0.97$; (3) Affect: mean=0.456, 89% CrI [0.019, 0.882], $P(>0)=0.95$; (4) Perception: mean=0.549, 89% CrI [0.114, 0.984], with a posterior probability positive of $P(>0)=0.98$; (5) Cognition: mean=0.497, 89% CrI [0.070, 0.938], with a posterior probability positive of $P(>0)=0.96$; (6) Volition: mean=0.404, 89% CrI [-0.019, 0.828], with a posterior probability positive of $P(>0)=0.94$ (Figure 2). A raw cohen's *d* has been reported for all subscales, indicating that the model is well specified (Table 1). Individual scores on each of the subscales are indicated in Supplemental Figure 2.

Anecdotally, participants reported less anxiety with pindolol compared to equi-intensive but higher pilot doses of DMT alone.

Cardiovascular effects

Mean heart rate increased for the DMT with placebo pre-treatment group by 13 bpm at 2 minutes compared to a 1 bpm increase

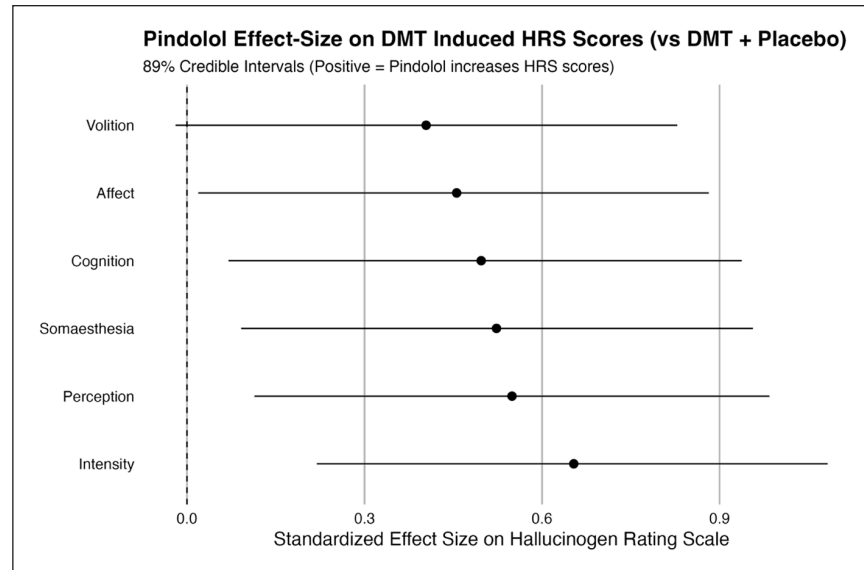


Figure 2. Individual subscale mean standardized effect.

The individual subscale-specific standardized effect size is plotted within an 89% credible interval.

Table 1. Cohen's *d* for all subscales.

Subscale	DMT + Placebo	DMT + Pindolol	Cohen's <i>d</i>
	Mean (SD) <i>N</i> = 12	Mean (SD) <i>N</i> = 12	
Intensity	1.62 (0.53)	2.04 (0.59)	0.74
Somaesthesia	0.72 (0.41)	0.99 (0.52)	0.57
Affect	1.00 (0.34)	1.25 (0.60)	0.50
Perception	1.24 (0.80)	1.64 (0.49)	0.61
Cognition	0.80 (0.57)	1.10 (0.56)	0.54
Volition	1.20 (0.66)	1.49 (0.63)	0.44

Sample size, mean, standard deviation, and Cohen's *d* for all subscales calculated using the raw values.

DMT: *N,N*-dimethyltryptamine.

for the DMT with pindolol pre-treatment group (Figure 3). Mean peak blood pressure increases (systolic/diastolic) from baseline were 19 mmHg/7 mmHg at 2 minutes for the DMT with placebo pre-treatment group and 36 mmHg/15 mmHg at 2 minutes in the DMT with pindolol pre-treatment group (Figure 4). Similarly, for mean arterial pressure (MAP), there was an 11 mmHg increase in MAP for the DMT with placebo pre-treatment group (Figure 5) and a 22 mmHg increase in MAP in the DMT with pindolol pre-treatment group.

It is noteworthy that blood pressure ranged from 117 mmHg/77 mmHg to 194 mmHg/109 mmHg in the DMT with pindolol pre-treatment group at 2 minutes. The participant who experienced the highest blood pressure peaked at 2 minutes (194 mmHg/109 mmHg) and lowered to a blood pressure of 139 mmHg/70 mmHg at 5 minutes (Supplemental Figure 2). Individual blood pressure, both absolute values and change from baseline, is reported for all drug groups in the supplement (Supplemental Figures 3–6).

Discussion

The present analysis is the first to quantitatively demonstrate that 5-HT_{1A} receptor blockade *increases* the subjective effects of DMT with a moderate effect size. Our analyses show a robust increase in global subjective effects as measured by the HRS. In the context of a previous report demonstrating that psilocybin plus pretreatment with the 5-HT_{1A} agonist buspirone *decreases* subjective effects (Pokorny et al., 2016), these data suggest functional antagonism between psychedelic-induced activity at 5-HT_{1A} receptors and 5-HT_{2A} receptors. It is also possible, though less likely, that there is an unknown pharmacokinetic interaction between pindolol and DMT such that it reduces DMT clearance.

We also find that pindolol pretreatment enhanced blood pressure response to DMT, in a way similar to that seen with the psychological effects. This suggests an inhibitory effect of 5-HT_{1A} receptor agonism on cardiovascular function, blockade of which by pindolol allows unopposed cardio-stimulatory 5-HT_{2A} receptor responses (McCall et al. 1987). Alternatively,

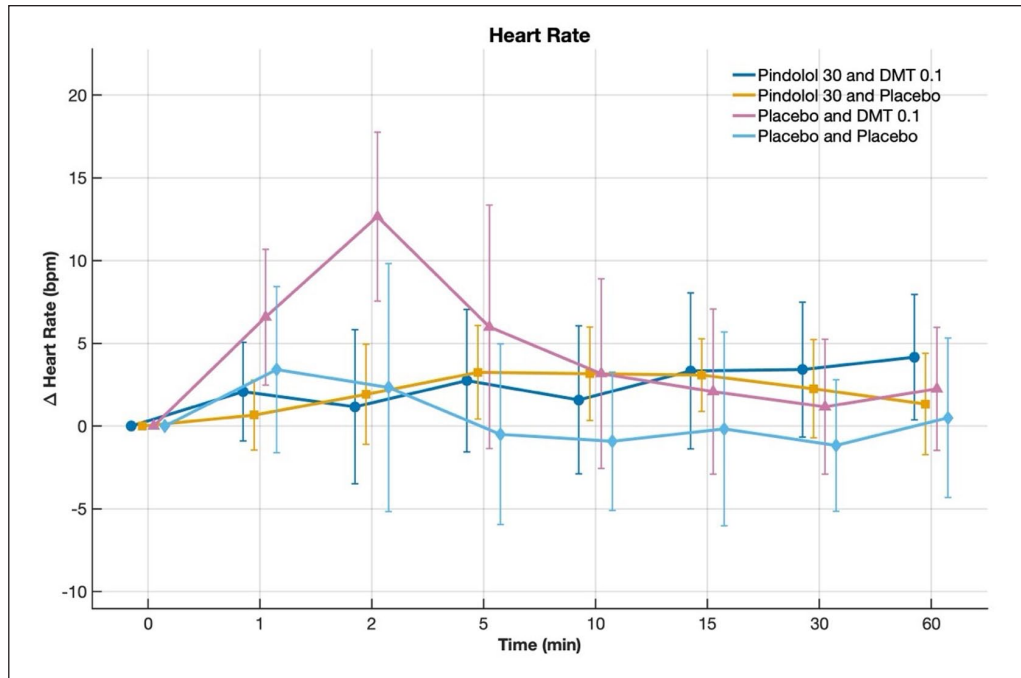


Figure 3. Mean change in heart rate from baseline following drug administration. Mean change in heart rate from baseline is plotted up to 60 minutes following drug administration in $N=12$ participants.

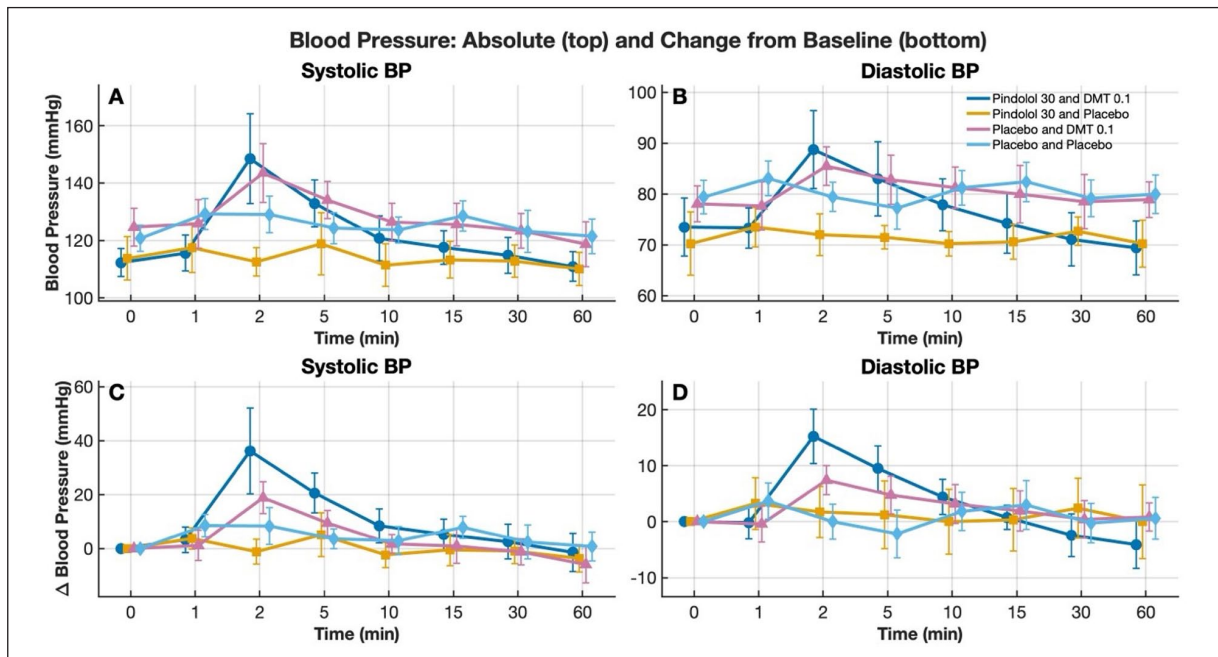


Figure 4. Changes in blood pressure following drug administration. (a and b) Absolute change in blood pressure and (c and d) Mean change in blood pressure from baseline is plotted up to 60 minutes following drug administration in $N=12$ participants.

this may simply be the product of intensified subjective effects and subsequent sympathetic nervous system activation.

5-HT_{1A} Receptors and 5-HT_{2A} receptors have opposing effects on neuronal circuitry. Enriched in the somato-dendritic compartments of pyramidal neurons, both receptors are found in

high density in cortical areas (Palchadhuri and Flügge, 2005; Weber and Andrade, 2010). The post-synaptic 5-HT_{1A} receptor also exists in limbic and midbrain regions in high density on cells spanning GABAergic interneurons, excitatory pyramidal neurons, and granule cells (Razakarivony et al., 2021; Palchadhuri

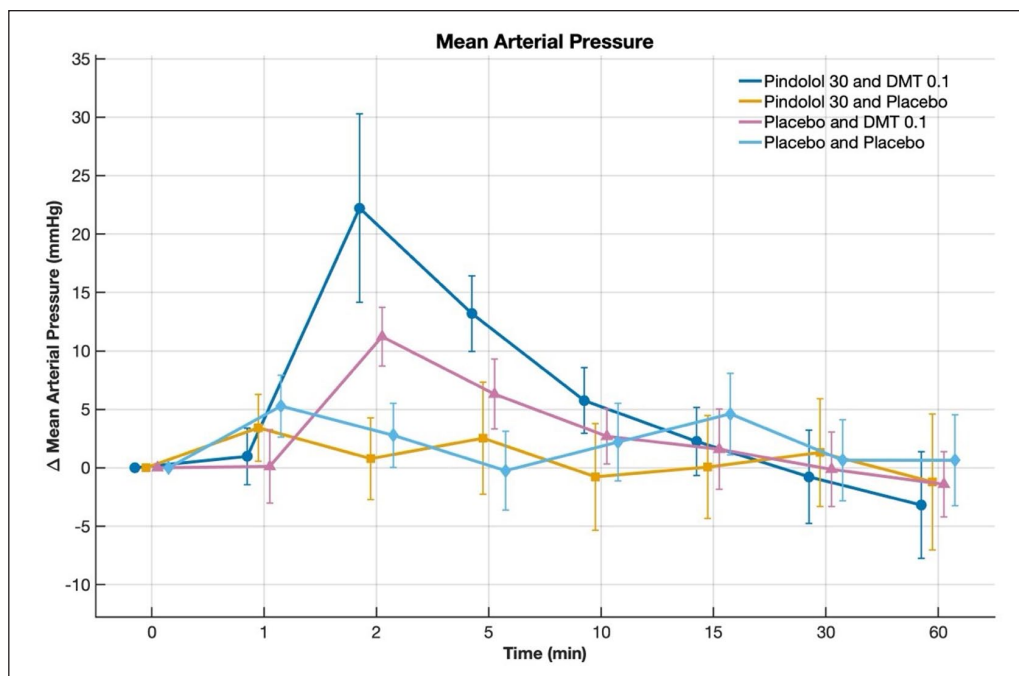


Figure 5. Mean change in mean arterial pressure from baseline following drug administration. Mean change in mean arterial pressure is plotted up to 60 minutes following drug administration in $N=12$ participants.

and Flügge, 2005), primarily modulating 5-HT transmission through hyperpolarization of the host neuron, thus causing inhibition of excitatory neurons and disinhibition of pyramidal neurons with 5-HT_{1A} receptor-containing interneuron afferents (Palchoudhuri and Flügge, 2005; Schmitz et al., 1998).

In contrast, 5-HT_{2A} receptors induce depolarization of the host neuron, primarily modulating 5-HT transmission through the excitation of excitatory pyramidal neurons in layer V of the cortex (Andrade, 2011). It is striking to note that 80% of pyramidal neurons in the prefrontal cortex co-express these neurons that induce opposing effects on the host neuron, thus making clear the complex interplay of electrophysiological properties these receptors confer on functional dynamics. Such interactions may significantly affect the excitation/inhibition (E/I) balance in the brain, a measure purported to contribute to the biological and psychological effects of serotonergic psychedelics, by shifting the psychedelic brain to a more globally excited state (Jardri et al., 2016; Lu et al., 2021). Thus, it follows that varying degrees of 5-HT_{2A} and 5-HT_{1A} receptor activation may produce distinct shifts in E/I balance, leading to correspondingly different alterations in perception. It is necessary to test changes in E/I balance during the acute experience in the presence and absence of 5-HT_{1A} agonism and antagonism in humans using an aperiodic analysis of electroencephalography (EEG) taken during the experience. These results would be strengthened by positron emission tomography (PET), measuring receptor occupancy of 5-HT_{1A} and 5-HT_{2A} receptors across a host of psychedelics. Such work would prove pivotal in our understanding of receptor-level contributions to the subjective experience.

Notably, pre-treatment of psilocybin with the 5-HT_{1A} agonist buspirone markedly reduces visual effects more than other subjective effects (Pokorny et al., 2016). Buspirone reduces “elementary” and “complex visual hallucinations” by half, representing the

largest attenuation across all subjective domains. This suggests that 5-HT_{1A} agonism may preferentially dampen visual components of the psychedelic experience.

It remains unclear whether 5-HT_{1A} blockade merely intensifies subjective effects globally or selectively alters some phenomenological features more than others. Notably, 5-MeO-DMT has much greater affinity for the 5-HT_{1A} receptor than other commonly used psychedelics (Ermakova et al., 2022) and appears to have distinct subjective effects. Most prominently, 5-MeO-DMT likely produces markedly fewer visual phenomena, and possibly produces more content-minimal phenomenology with higher rates of ego dissolution (Blackburne et al., 2025; Davis et al., 2018; Ermakova et al., 2022; Timmermann et al., 2025). However, to date, there has been no work comparing 5-MeO-DMT to other psychedelics.

We hypothesize that the unique phenomenological effects of 5-MeO-DMT relative to other psychedelics result from engagement of the 5-HT_{1A} receptor. Accordingly, we anticipate that blocking this receptor with pindolol would modify the subjective effects of 5-MeO-DMT, potentially shifting the experience toward a more perceptually elaborated or visual state.

We assume that pindolol pre-treatment of psilocybin would similarly intensify subjective effects. The use of more granular, validated subjective measures may allow detection of preferential changes in particular phenomenological dimensions, as opposed to a general increase across all subjective effects. Anecdotal reports from participants that pindolol reduced anxiety, compared to previous high doses of DMT, are consistent with the known anxiolytic effects of β -blockade and justify further evaluation of this approach in psilocybin therapy.

Our work was limited by the use of a single survey, the HRS, to measure subjective outcomes, as well as a single dose for both DMT and pindolol. In future work, it is vital to use additional

measures of subjective experience and varied doses of both the psychedelic and pindolol to assess the full range of effects of 5-HT_{1A} blockade.

Overall, this work reinforces the importance of 5-HT_{1A} signaling in the acute subjective effects of serotonergic psychedelics and points toward clear, testable hypotheses for future work with psilocybin and 5-MeO-DMT.

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Declaration of conflicting interests

SMN has consulted for Lykos Therapeutics, AbbVie, and MMS Holdings.

Supplemental material

Supplemental material for this article is available online.

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